

DART: From Attentive Reasoning to Intuitive Execution for Experience-Driven Robot Intelligence

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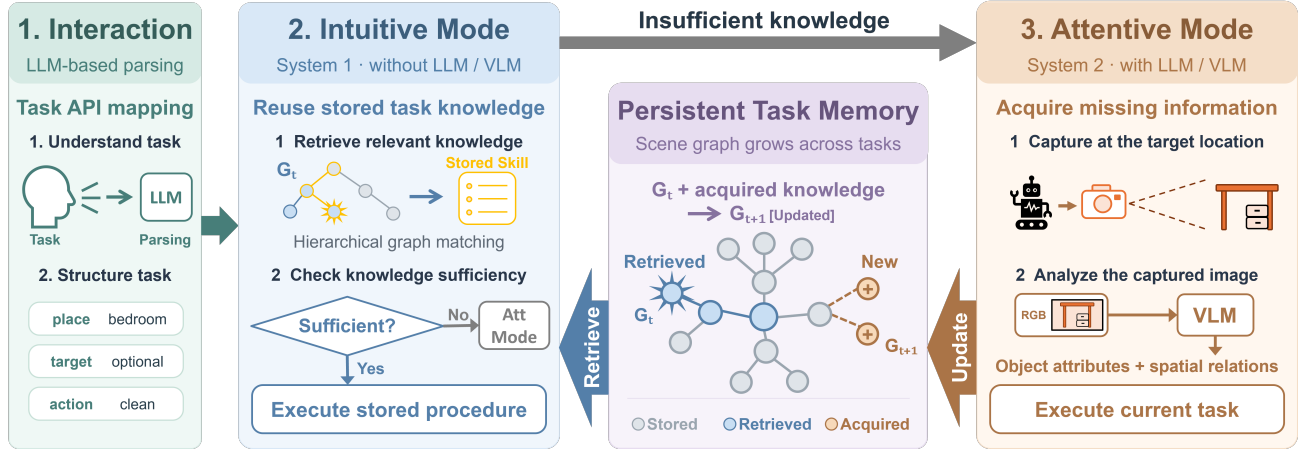


Fig. 1. Overview of DART, a dual-mode architecture for robot intelligence. *Interaction layer* parses tasks and structures them into a memory-aligned format. *Intuitive Mode* searches persistent task memory for relevant knowledge and directly executes the task using the retrieved information. When the available knowledge is insufficient, *Attentive Mode* selectively gathers functional information needed for the task and executes it. The acquired information is retained in a persistent scene graph, enabling DART to rely more on *Intuitive Mode* for later related tasks with reduced reliance on high-capacity models.

Abstract—Humans rely on both intuitive and deliberate processing, as described by dual-process theories of cognition: novel tasks often require effortful reasoning, whereas accumulated experience allows similar tasks to be handled more automatically and with less cognitive effort. Motivated by this principle, we propose DART (Deliberation-Adaptive Reasoning from Task Experience), which applies the same principle to robots through two connected modes. When the robot lacks sufficient task-relevant knowledge or prior experience, it operates in Attentive Mode, performing high-level reasoning and acquiring the additional knowledge needed for the task. The newly acquired knowledge is stored in a persistent scene graph and accumulated across tasks. As experience builds, the robot increasingly operates in Intuitive Mode, reusing this knowledge for later related tasks without repeated high-level reasoning. We evaluate DART in simulated and real-world environments, including deployment on a resource-constrained platform, and show that as knowledge accumulates in the scene graph, the robot requires less high-level reasoning and less additional information gathering while maintaining task performance.

I. INTRODUCTION

Humans and animals can learn from a handful of experiences. Equipped with basic motor skills and common-sense knowledge, they can sometimes figure out novel tasks after only a few trials, and once those tasks become routine, executing them becomes effortless. We believe truly intelligent robots should be able to do the same.

Let's imagine a home robot deployed in a user's house for a long period. We observe that while the variations in all possible houses are enormous, the variations in a particular house may be small. That is, it may not be able to anticipate

the environment and the tasks from the user, but it may be able to get used to the majority of the routine tasks quickly. We also observe that many household tasks, such as cleaning, navigation, and object retrieval, are associated with a subset of the semantic knowledge available in the environment. Across these tasks, a small set of semantic concepts is often reused, including important locations and objects, and the spatial relations among them. Such information can be represented using 3D semantic scene graphs [1]–[3]. However, constructing semantic representations in advance [1]–[4] may miss semantic information that is relevant to the user [5], whereas task-time semantic reasoning may incur computational overhead through repeated invocations of high-capacity models [6]–[8].

Here, we propose **DART (Deliberation-Adaptive Reasoning from Task Experience)**, an architecture for robot intelligence motivated by dual-process theories of cognition [9], [10]: novel tasks often require high-capacity, slow thinking (System 2), whereas routine tasks are handled by fast thinking that requires substantially less cognitive effort (System 1).

In DART, we take the view that the two systems operate at the same level, and that only one of the systems is generating an output at any given time: System 1 is active for routine tasks that the system already understands, and System 2 takes over when the task is unfamiliar or when System 1 fails. Novel tasks solved by System 2 become part of System 1, as successful experience accumulates in the structured memory.

More specifically, DART operates in two complementary

modes: In **Attentive Mode**, the robot reasons about the task, determines what additional knowledge is needed, and selectively acquires task-relevant information during execution. Knowledge acquired in Attentive Mode is retained across tasks and can reduce the need for attentive reasoning in later related tasks. In **Intuitive Mode**, the robot uses structured knowledge already available to it, reducing the need for repeated reasoning.

We show a minimal instance of DART on a resource-constrained, consumer-grade robot vacuum using a compact, persistent scene graph which encodes task-relevant information such as viewpoints for important places and objects. During early task executions, the robot acquires task-relevant information from stored observation viewpoints using a cloud-hosted vision-language model (VLM) in Attentive Mode, and the acquired information is integrated into the persistent graph and becomes available for subsequent executions via Intuitive Mode.

The main contributions of this work are as follows:

- We introduce DART, a dual-mode architecture for robot intelligence that integrates attentive reasoning with increasingly intuitive execution of related tasks as experience accumulates, without updating neural-network parameters during task execution.
- We instantiate DART on a robot vacuum using a compact, persistent scene graph, lightweight on-device perception, and task-driven semantic acquisition through language and vision-language models.
- We evaluate DART in simulated and real-world environments, showing fewer Attentive Mode invocations and less additional information gathering as knowledge accumulates while maintaining task performance.

II. RELATED WORK

A. Semantic Mapping and Structured Scene Representations

Semantic mapping combines geometric maps with semantic information about objects and spatial regions. Such representations include dense metric maps, object-centric maps, and graph-based structures. Dense methods [11] fuse semantic predictions from multiple observations into geometric reconstructions, whereas other approaches explicitly model objects or scene components using volumetric, topological, or object-graph structures [12]–[14].

Scene graphs represent an environment using semantic entities such as objects, places, and regions together with their spatial relations [15]. Kimera and Hydra [1], [2], [16] automate 3D scene-graph construction from robot observations. Khronos [17] models short-term dynamics and long-term environmental changes within a spatio-temporal metric-semantic map. Recent methods [18], [19] use foundation models to construct open-vocabulary scene graphs or associate spatially grounded language descriptions with scene representations. FunGraph [20] extends 3D scene graphs with functional information for language-guided interaction. In this work, we use a compact scene graph as a persistent representation of the environment. Task-relevant semantic

information acquired in Attentive Mode is retained in the graph and reused for later related tasks.

B. Task-Driven Semantic Acquisition and Active Perception

Semantic representations have been used to support robotic task execution, including task planning and object search. SayPlan [21] uses a large language model (LLM) to identify relevant portions of a pre-built hierarchical 3D scene graph for task planning. Other approaches [6], [22] construct and update scene graphs during exploration and use language models for planning and reasoning.

Active perception methods guide exploration to acquire task-relevant observations. For example, GenMOS [23] maintains octree-based beliefs over object locations and selects viewpoints via online planning. Other VLM-based methods use semantic information to guide exploration [7], [8], while AP-VLM [24] uses a 3D virtual grid and a VLM to select viewpoints. VISTA [25] plans task-relevant exploration trajectories while building a real-time semantic 3D reconstruction. When existing knowledge is insufficient, DART revisits precomputed observation viewpoints to selectively acquire additional task-relevant semantic information. The acquired knowledge is stored in the persistent scene graph and can reduce the need for high-level reasoning in later related tasks.

C. Resource-Efficient Semantic Processing and Dual-Process Architectures

Several studies have investigated efficient semantic processing in robotics. HOV-SG [3] represents scenes at floor, room, and object levels, while Clio [26] uses the information bottleneck principle to determine task-dependent scene granularity. FOUND-IT [27] resolves object instances on demand after receiving a query, enabling task-dependent scene granularity. The method in [28] targets real-time scene-graph construction using CPU-only processing, whereas GraphPad [5] and JITOMA [29] selectively update or expand structured memory based on the current task. EfficientEQA [30] and ObsGraph [31] guide exploration using semantic relevance or evidence sufficiency.

Dual-process architectures distinguish between fast and slow processing modes. Helix [32] and Hi Robot [33] combine slower high-level semantic processing with faster low-level visuomotor control, separating the two processes across different levels of abstraction and control. IROS [34] invokes VLM reasoning when fast condition matching is ambiguous, while CogDDN [35] uses an analytic process to revisit failures and retain corrected experience. FSR-VLN [36] uses a hierarchical scene graph for an initial candidate search and reserves VLM reasoning for cases requiring further disambiguation, whereas INHerit-SG [37] places geometric graph updates and semantic interpretation on separate asynchronous paths during incremental mapping. DART instead applies the dual-process distinction directly to semantic reasoning: Intuitive Mode reuses semantic knowledge accumulated across related tasks, whereas Attentive

Mode performs additional deliberation and information acquisition when that knowledge is insufficient.

III. PROBLEM FORMULATION

We consider repeated task execution in the same or slowly changing environment, with the goal of reusing accumulated knowledge and acquiring additional task-relevant information only when the available knowledge is insufficient. In our system, DART is instantiated on a resource-constrained robot using a scene graph for task execution. The robot is equipped with a 2D LiDAR, an RGB-D camera, odometry, and a lightweight closed-vocabulary object detector, and can access language and vision-language models as additional knowledge sources. Given a natural-language task q and the current graph $\mathcal{G}_t$, DART generates and executes an action sequence.

DART represents its persistent task memory as a scene graph $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}, \mathcal{X})$, where $\mathcal{X}$ stores geometric and semantic attributes. The node set is $\mathcal{V} = \mathcal{V}_R \cup \mathcal{V}_S \cup \mathcal{V}_O \cup \mathcal{V}_P$, representing regions, semantic subspaces, objects, and observation viewpoints, respectively. The edge set is

$$\mathcal{E} = \mathcal{E}_{RR} \cup \mathcal{E}_{RS} \cup \mathcal{E}_{RO} \cup \mathcal{E}_{RP} \cup \mathcal{E}_{OP} \cup \mathcal{E}_{sem}, \quad (1)$$

where $\mathcal{E}_{RR}$ encodes region connectivity, $\mathcal{E}_{RS}$ region-subspace containment, and $\mathcal{E}_{RO}$ region-object associations. $\mathcal{E}_{RP}$ and $\mathcal{E}_{OP}$ link regions and objects to their observation viewpoints, respectively. $\mathcal{E}_{sem}$ contains task-acquired semantic relations, including object-object relations. The graph is initialized as $\mathcal{G}_0$ through lightweight on-device perception and mapping and updated with knowledge acquired during task execution.

Given q , Intuitive Mode executes the task when the knowledge in $\mathcal{G}_t$ is sufficient. Otherwise, Attentive Mode acquires the missing knowledge and executes the task. The acquired information is retained in $\mathcal{G}_{t+1}$ for reuse in subsequent tasks.

IV. METHOD

A. Framework Overview

DART consists of an interaction layer, Intuitive Mode, and Attentive Mode sharing a persistent functional scene graph initialized through lightweight on-device mapping and perception. The interaction layer structures task inputs. Intuitive Mode uses stored knowledge for execution, while Attentive Mode acquires and retains missing knowledge for current and subsequent tasks.

B. Initial Lightweight Scene Construction

DART constructs an initial scene representation from an occupancy grid map generated by 2D LiDAR SLAM and on-device closed-vocabulary object detections obtained from RGB images during navigation. As shown in Fig. 2, the resulting geometric and semantic elements are organized according to a predefined graph schema.

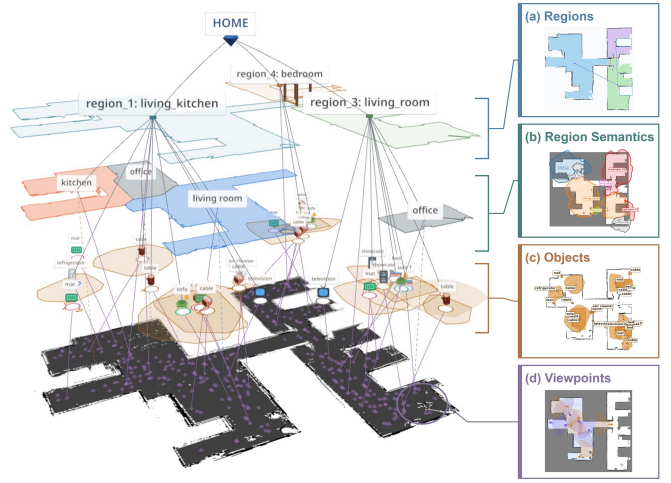


Fig. 2. Left: the initial scene graph integrating geometric regions, semantic subspaces, objects, and observation viewpoints. Right: (a) geometric region segmentation, (b) region classification and semantic subspace extraction, (c) multi-view object mapping, and (d) observation viewpoint generation.

1) *Geometric Region Segmentation*: After navigation and SLAM are completed, geometric region segmentation is performed on the occupancy grid map. Using only map geometry, free space is partitioned by a distance-transform-based method, and adjacent spaces are separated at narrow passages detected as doorways. Regions that are excessively small or large are merged or split by size. Corridor-like structures and bed-like obstacle patterns are also extracted as geometric cues. When a geometric cue is associated with an object, the corresponding object detection is used to verify the cue and update its geometric extent.

2) *Semantic Region Classification*: The type of each segmented region is estimated from object observations accumulated during navigation. These observations are obtained using a YOLOv8-based detector [38] retrained for the target indoor object classes. Detection results are post-processed to obtain image-plane object contours. Inspired by BORM [39], the region classification process uses object occurrence likelihoods across indoor space categories as semantic evidence.

DART uses the occupancy grid map and the camera pose of each observation. For object class o and region type c , a predefined likelihood $\lambda(o, c)$ is used as semantic evidence. Given a detection d , the accumulated score $S_c(x)$ is updated over the map area Ω_d covered by the camera's field of view (FOV) as $S_c(x) \leftarrow S_c(x) + \lambda(o_d, c)$ for $x \in \Omega_d$, where o_d is the detected class and x is a 2D map location. For each map cell, the region type with the highest accumulated score is selected. Within each geometric region, the region type assigned to the largest area is used as the representative label, while the remaining class-specific areas are retained as subspaces. Each subspace is represented by a 2D polygon constructed from the map cells assigned to that semantic area.

3) *Multi-View Object Mapping*: The same detection results are also used for multi-view object mapping, which estimates the 2D location and spatial extent of each object.

We assume that the target objects are floor-supported. The image-plane contour of each detection is projected onto the ground plane using the camera model and camera pose to obtain an initial 2D footprint. Depth measurements are then used to constrain the projection to the observed object extent. The initial scene graph represents each object by its class, 2D map location, and planar extent. We therefore integrate object observations directly in the 2D map without per-object 3D reconstruction or object pose tracking.

Observations of the same class are associated based on footprint overlap or center distance; unmatched observations create new objects. Associated observations that do not expand the current footprint only increment the observation count. After outliers are removed from the accumulated footprint vertices, the convex hull is updated, and the mean of its vertices defines the representative map location.

4) *Observation Viewpoints*: Observation viewpoints are constructed for regions and objects to support semantic information acquisition in future tasks. Candidate viewpoints are associated with objects and regions based on observations and robot poses collected during mapping, with visibility and accessibility considered in their selection. Each viewpoint specifies a map position and an observation yaw. Task-dependent viewpoint selection and semantic acquisition are described in Sec. IV-E.

5) *Initial Scene Graph Assembly*: Finally, the outputs of region segmentation, region classification, object mapping, and viewpoint selection are integrated into the initial scene graph. During mapping, DART does not continuously maintain or update the scene graph; it maintains the occupancy map and accumulates object observations instead. After mapping, the lightweight module outputs are assembled into a compact graph.

Each geometric region forms a region node, with secondary subspaces as child nodes. Objects are linked to their parent regions based on 2D map location, forming region-object relations. Region viewpoints are stored as candidate poses for acquiring additional semantic information about the corresponding space, while object viewpoints support further observation of specific objects. Adjacent-region connectivity is defined using detected doorways. Semantic information and relations acquired during task execution are added to the same persistent graph.

C. Interaction Layer: Task Parsing and Structured Representation

The interaction layer parses task inputs in different formats into a structured representation that enables hierarchical search and matching within the functional scene graph. An LLM extracts the requested task and relevant area, region, object, and description information, organizing them into fields aligned with the graph hierarchy. The resulting representation is passed to Intuitive Mode for graph search.

D. Intuitive Mode: Graph Search and Graph-Grounded Execution

Intuitive Mode uses the structured task representation to search the persistent functional scene graph without invoking

an LLM. It executes the task when the required knowledge is available; otherwise, it passes the task and the identified semantic gap to Attentive Mode.

1) *Task Analysis and Graph-Grounded Knowledge Retrieval*: The scene graph follows a space-region-object hierarchy, along which Intuitive Mode searches task-relevant nodes and edges. The structured task fields are progressively matched against the corresponding graph entities and their stored descriptions, narrowing the search to entities that satisfy the specified conditions. From the matched nodes and edges, DART retrieves task-relevant graph context, including object positions, region geometry, region connectivity, semantic and functional relations, and stored actions and execution procedures.

Object-related tasks are grounded to an object node and its available spatial information, whereas space-related tasks are grounded to a region or subspace node. For an object-related task, DART retrieves the object node, its stored location when available, and the space or region associated with the object. For a space-related task, it retrieves the corresponding region or subspace node and its stored geometry. The requested task is then matched against the functional relations and execution knowledge associated with the retrieved entities to identify the corresponding actions and procedures.

2) *Direct Task Execution*: After retrieving the task-relevant graph context, DART checks whether the specified task fields have been matched and the corresponding spatial information, actions, and execution procedures are available. When the required knowledge is available, DART directly executes the task by following the procedures stored in the graph.

For an object-related task, the stored object location and its associated space provide the spatial reference for execution. For a space-related task, the geometry and location of the corresponding region or subspace are used. The retrieved execution procedures specify the actions and task-specific settings to apply to the grounded target. These procedures are carried out through the robot's available capabilities, with navigation supported by the occupancy map when movement is required.

If information required for execution is unavailable, DART defines the missing information as a semantic gap and passes the task, retrieved graph context, and semantic gap to Attentive Mode.

E. Attentive Mode: On-Demand Semantic Acquisition

Attentive Mode selectively acquires the knowledge needed to resolve the semantic gap identified by Intuitive Mode. It receives the structured task representation, the retrieved graph context, and the failure type and stage at which matching or retrieval failed.

Based on the semantic gap, DART uses an LLM to reason about missing functional or execution knowledge and a VLM to acquire task-relevant information about objects and spaces. Both processes use the current functional scene graph as shared task context. Attentive Mode executes the task using

the retrieved and newly acquired knowledge, then retains all newly acquired knowledge in the persistent graph.

1) *Predefined Observation Viewpoints*: To acquire missing task-relevant information, DART selects a stored observation viewpoint associated with the target object or region from the initial scene graph. The selected viewpoint provides an initial observation location for semantic acquisition.

2) *Targeted Semantic Acquisition*: Given the semantic gap identified by Intuitive Mode, Attentive Mode determines what additional knowledge is needed using the structured task representation, the retrieved graph context, and the reported failure type. When the gap concerns task understanding, functional relations, or execution procedures, an LLM queries the current scene graph and reasons about how the requested task can be fulfilled. The resulting knowledge specifies relevant functions, suitable target entities, or actions and execution procedures grounded in the available graph context and the robot’s capabilities.

When the gap concerns an object or space that requires visual observation, DART determines an acquisition target. If the requested entity is not represented in the current scene graph, the LLM queries the graph and combines the retrieved spatial and semantic context with common-sense knowledge and task-specific rules to infer candidate spaces in which the entity is likely to be found. The identified object or space is used as the acquisition target.

For visual information acquisition, DART uses a predefined viewpoint associated with the acquisition target as an initial observation location. The robot navigates to the viewpoint and captures an RGB image, which is provided to the VLM together with a task-specific query derived from the semantic gap.

The VLM produces task-relevant object attributes and descriptions, space–object associations, object–object relations, and space-related semantic information. The acquired results are added to the current task context, after which DART determines whether sufficient knowledge is available to complete the task. Additional observations or LLM-based reasoning are used as needed to resolve the remaining gap.

Once sufficient knowledge is available, Attentive Mode executes the task using the spatial information, actions, and execution procedures retrieved from the graph or established during knowledge acquisition. The robot carries out these procedures through its available capabilities.

3) *Semantic Knowledge Integration*: The knowledge acquired in Attentive Mode is used for executing the current task. Upon task completion, DART integrates all acquired knowledge into the persistent functional scene graph. This includes semantic and functional information as well as actions and execution procedures, making them available for reuse in subsequent tasks.

Information about an existing object is added to the corresponding node as task-relevant attributes and semantic descriptions. A newly identified object is represented as a new object node and connected to the region or space in which it was observed. Spatial information obtained through the VLM is represented as a region-level association.



Fig. 3. Illustrative examples of the experimental environments.

If this association is insufficient for the requested task, DART acquires the additional spatial information needed for execution in Attentive Mode. Object–object relations and object–space associations are represented as graph edges, with their relation types and semantic descriptions stored as edge attributes.

Space-related semantic information extends the descriptions of existing space or region nodes and introduces semantic relations between the associated entities as graph edges. Functional knowledge associates objects and spaces with the functions they support. Actions, task-specific settings, and execution procedures used to complete the task are stored in association with the corresponding entities and task specifications.

V. EXPERIMENTS

A. Implementation Details

For the experiments, we used a commercial robot vacuum cleaner to evaluate DART on a resource-constrained mobile robot. The robot has a maximum translational speed of 0.3 m/s. It is equipped with a 2D LiDAR, an RGB camera, and a short-range depth camera. The RGB camera provides images at 800×1280 pixels, while depth images are processed at 216×136 pixels. The robot uses an in-house 2D LiDAR-based SLAM system for mapping and localization and a grid-based navigation system.

For simulation, we modeled the robot in NVIDIA Isaac Sim 5.1.0¹ to reflect the physical robot’s dimensions, motion limits, and sensor configuration. The simulated robot used Cartographer [40] for mapping and localization, and ROS 2 Nav2 [41] for navigation.

In both simulated and real-world experiments, we used YOLOv8 retrained to detect common household objects in images captured near floor level. Qwen3.5 [42] was used for instruction parsing in the interaction layer and for text-based reasoning and visual semantic acquisition in Attentive Mode. Qwen3.5 inference was performed on an external server.

B. Experimental Environments and Datasets

We used two residential environments in our experiments: Simulated Home (Sim-Home) and Real Home (Real-Home).

¹<https://docs.isaacsim.omniverse.nvidia.com/5.1.0/index.html>

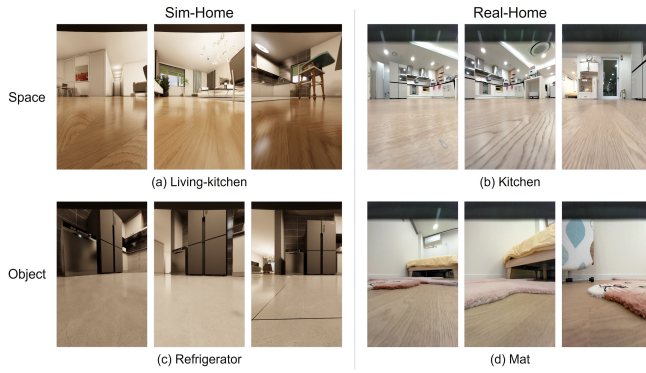


Fig. 4. Object and space observation viewpoints in Sim-Home and Real-Home: (a) living-kitchen, (b) kitchen, (c) refrigerator, and (d) mat.

Region segmentation was additionally evaluated using a real-world LiDAR maps dataset. Fig. 3 shows the Sim-Home and Real-Home environments.

Task-Execution Environments. Sim-Home is a custom-built simulated residential environment modeled at real-world scale. It comprises a large open-plan area combining the kitchen, living room, and dining area, along with two bedrooms, one of which has an en-suite bathroom. Real-Home is a real-world residential environment comprising an open-plan area that combines the living room, dining area, and kitchen, along with one bedroom.

Initial Scene-Graph Data Collection. In both environments, the robot followed a zigzag trajectory over a regular grid covering the accessible areas. RGB images, robot poses, and object detections were recorded every 1 m of travel in Sim-Home and at 10 Hz in Real-Home. These data were used to construct an initial scene graph and establish observation viewpoints for each environment.

Residential LiDAR Map Dataset. This dataset consists of 159 2D LiDAR-based maps collected in real-world residential environments. Ground-truth regions were manually annotated for evaluating geometric region segmentation.

C. Component-Level Evaluation of Scene Graph Construction

We evaluate four aspects of initial scene-graph construction: region segmentation, object mapping, region classification, and observation viewpoint reliability. Region segmentation was evaluated using the Residential LiDAR Map Dataset. The method achieves 57.96% precision, 68.17% recall, and 62.65% F_1 (Table I). The mean intersection over union (IoU) over valid matched region pairs is 85.10%; unmatched regions are not included in this average. Object mapping is evaluated in Sim-Home using precision, recall, F_1 , localization error, and footprint IoU. A predicted object is matched to a ground-truth object of the same class when the distance between their centers is at most 0.5 m. Localization error and footprint IoU measure the accuracy of the estimated 2D object location and planar extent, respectively (Table II). Region classification is evaluated on Sim-Home and Real-Home using accuracy and macro- F_1 . Macro- F_1 is

TABLE I

PERFORMANCE OF THE REGION SEGMENTATION ALGORITHM ON THE RESIDENTIAL LiDAR MAP DATASET.

Precision (%)	Recall (%)	F_1 (%)	IoU (%)
57.96	68.17	62.65	85.10

TABLE II

PERFORMANCE OF THE OBJECT MAPPING ALGORITHM ON SIM-HOME.

Precision (%)	Recall (%)	F_1 (%)	Loc. Error (m)	IoU (%)
50.00	82.00	62.12	0.23	42.60

the unweighted mean of the class-wise F_1 scores. When the evaluated regions from Sim-Home and Real-Home are pooled, the method achieves 83.33% accuracy and 86.67% macro- F_1 . Observation viewpoint reliability is evaluated by checking whether a human evaluator can identify the predefined target space or object in the image associated with each stored viewpoint. Only image–target pairs with a definite success or failure judgment are included in the calculation. The success rates are 95.18% (237/249) in Sim-Home and 88.03% (103/117) in Real-Home (Table III). Fig. 4 shows example observations of an object and a space from their stored viewpoints.

D. Task Execution Performance

We evaluate whether retaining task-acquired knowledge reduces VLM usage and improves task success compared with discarding that knowledge after each execution. For each of two environments—Sim-Home and Real-Home—we construct 20 tasks comprising object search and navigation to specified locations. Each task execution starts from the same pose in front of the docking station and is limited to 20 VLM calls in total. We count only VLM tokens and calls for visual information acquisition and description generation, with each inference request counted as one call. Task success is assessed from execution logs and final responses based on whether the task instruction is satisfied. We compare DART with DART-Frozen, which also uses VLM-based observations to acquire information for the current task but does not retain the acquired knowledge for subsequent executions.

Single-Task Repetition. To examine knowledge reuse within the same task, we execute one task ten times consecutively, retaining acquired knowledge across repetitions. We then restore the scene graph to its initial state before repeating the next task ten times. Fig. 5(a) shows a sharp, sustained reduction in VLM token usage after the first repetition in both environments, with each point averaged across 20 tasks. Mean token usage remains below the first-execution level throughout subsequent repetitions, although it fluctuates. This result is consistent with acquired knowledge reducing the need for repeated visual information acquisition when the same task is executed again.

Task-Sequence Repetition. To evaluate knowledge reuse across tasks and iterations, the robot executes the tasks in

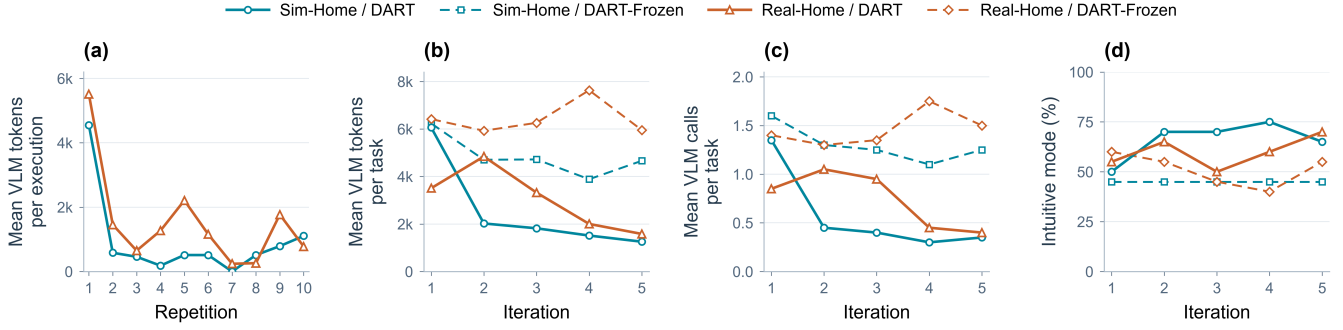


Fig. 5. Knowledge reuse and task-execution trends in Sim-Home and Real-Home. (a) Mean VLM token usage across repetitions, averaged over 20 tasks that are each repeated ten times. Knowledge is retained across the ten repetitions of each task, and the scene graph is restored to the same pre-update initial state before starting the next task. (b) Mean VLM tokens per task, (c) mean VLM calls per task, and (d) Intuitive Mode usage over five iterations of the 20-task sequence.

TABLE III

RE-OBSERVATION SUCCESS RATES AT STORED SPACE AND OBJECT VIEWPOINTS.

Dataset	Target	# Trials	# Successes	Success (%)
Sim-Home	Space	68	60	88.24
	Object	181	177	97.79
Real-Home	Space	60	51	85.00
	Object	57	52	91.23
Overall		366	340	92.90

TABLE IV

TASK SUCCESS RATES IN SIMULATION AND REAL-WORLD ENVIRONMENTS.

Dataset	Method	Success rate (%) $\uparrow$		
		Overall	Init.	Final
Sim-Home (Sim.)	DART-Frozen	55.0	55.0	50.0
	DART (Ours)	58.0	50.0	60.0
Real-Home (Real)	DART-Frozen	66.0	70.0	70.0
	DART (Ours)	87.0	90.0	90.0

Overall: iterations 1–5. Init.: iteration 1. Final: iteration 5.

order and repeats the complete sequence for 5 iterations. The sequence includes tasks with overlapping information requirements, allowing knowledge acquired during one task to support subsequent, distinct tasks. DART starts from the initial scene graph and retains acquired knowledge across tasks and iterations.

Table IV summarizes success rates over 5 iterations. DART achieves overall success rates of 58.0% in Sim-Home and 87.0% in Real-Home, exceeding DART-Frozen in both environments. For DART, final success rates are maintained or improved relative to the initial iteration. The results suggest that information acquired during earlier tasks helps DART complete subsequent tasks with overlapping information requirements.

Fig. 5(b)–(d) shows mean VLM tokens per task, mean VLM calls per task, and Intuitive Mode usage in both environments. Averaged over 5 iterations, DART uses 47.6%

fewer VLM tokens and 56.2% fewer VLM calls than DART-Frozen in Sim-Home; the corresponding reductions in Real-Home are 52.6% and 49.3%, respectively. Intuitive Mode usage increases from the initial to the final iteration. Together, these results show that DART achieves higher task success rates with fewer VLM calls and lower token usage than DART-Frozen. By retaining information acquired in Attentive Mode, DART makes it available to subsequent tasks with overlapping information requirements. The increase in Intuitive Mode usage further supports DART’s intended transition from acquiring missing information to reusing accumulated knowledge. These results demonstrate the feasibility of applying principles inspired by human cognition to robot intelligence by connecting attentive knowledge acquisition with intuitive knowledge reuse.

VI. CONCLUSION

In this work, we presented DART, a framework for robot intelligence in which robots use attentive reasoning to acquire task-relevant knowledge and later reuse the accumulated knowledge intuitively for related tasks. DART retains acquired knowledge in a compact, persistent scene graph, reducing the need for repeated high-level reasoning on later related tasks. Experiments in both simulated and real-world environments showed that DART generally uses fewer VLM calls and tokens as knowledge accumulates, while achieving higher overall task success than DART-Frozen. This reduced reliance on VLM inference supports DART’s deployment on a resource-constrained robotic platform. Future work will extend DART to more complex robotic platforms with manipulation capabilities, whose actions can alter object states and spatial relations. We will incorporate temporal information into the scene graph to track changes in object states and the environment over time, including those caused by the robot’s actions. We will also evaluate the framework through long-term deployments across more diverse environments and users.

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